

CIVIC AI INFRASTRUCTURE

THAT CAN BE QUESTIONED, EXITED AND IMPROVED

One civic spine, three Proof Stations, two service wings, six cross-connections and twelve governed scenarios translate the century-old railway grammar into public validation.

1141.3 ha

provisional area

18.0%

conceptual green

12 / 3

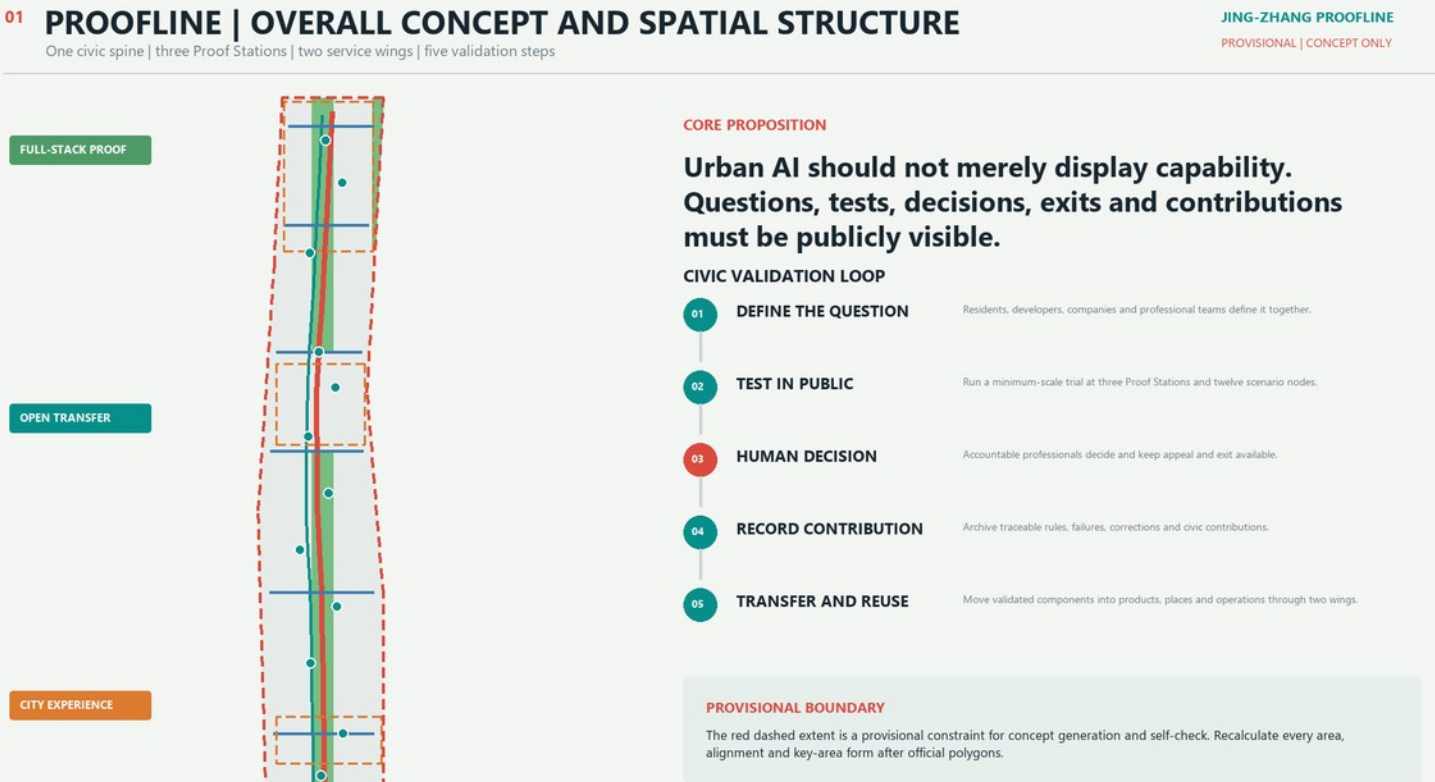
scenarios / industry tests

8

action packages

BOUNDARY STATUS

Provisional constraint for concept generation, display and self-check. It is not an Official Planning Boundary. Recalculate all layers, metrics, figures and drawings after rights-cleared official polygons.



EVIDENCE LEDGER UPDATE | HOW TRUSTED DATA ENTERS DESIGN

Clear provenance is sufficient for use only when license, statistical scale, purpose and limits are also registered. Background facts are not corridor measurements; catalog leads are not retrieved datasets.

01 | OFFICIAL ADMINISTRATIVE BASELINE

BEIJING 2025

209 filed and launched large models
180.7 high-value invention patents per 10,000 people
30 rail lines, 909 km; 2.74 billion express parcels

Calibrates the innovation, mobility and last-mile problem list; no demand is allocated to the Jing-Zhang corridor.

02 | OPEN-DATA CATALOG

60 SCHOOL-BUS CATALOG RECORDS

Fields list route, origin, destination, school, boarding place and departure time
Catalog says unconditional open; file link currently redirects to unified login

Register as a lead. Join aggregated records to stations and schools only after the file, coverage, time and license are verified.

03 | INDEPENDENT SPATIAL CHECK

0% INTERSECTION | 412.5 m NEAREST

17.49 ha mapped heritage park in OSM
100% covered by the research scope
Independent EPSG:4548 rerun matches Issue #846

Location-discrepancy risk only. Crowdsourced OSM is not a statutory redline and does not move PROV-SITE-001.

THREE USE CLASSES | NO CATEGORY LEAP

BACKGROUND FACT

Explains scale and context
Never enters corridor metrics

CATALOG LEAD

Registers fields and access path
No count claim before retrieval

DISCREPANCY FLAG

Exposes provisional-boundary risk
Official polygon adjudicates

DESIGN DECISION IN THIS PASS

CHANGED | Evidence explanations in sources, assumptions, risk, bilingual reports, HTML and formal drawings

UNCHANGED | GeoJSON, metrics, spatial placement, quantities, action packages and phasing

USE RULE: CLEAR PROVENANCE + COMPLIANT LICENSE + FIT-FOR-PURPOSE USE + VISIBLE LIMITS

Sources: Beijing 2025 Statistical Bulletin | Beijing Open Data catalog 'Haidian school-bus route information' | OpenStreetMap/Overpass (ODbL), Issue #846 / merged PR #850

02 DESIGN BASIS AND SOURCE TIERS

Use each source only for the claim it can support

OFFICIAL TASK BASIS

Announcement and local standards support tasks, scope text, approximate area and professional principles - not exact polygons or controls.

RIGHTS-CLEARED TASKBOOK

Supports brand, cases, scenarios, landmarks, culture and operations - not statutory planning or investment commitments.

PROVISIONAL SPATIAL BASIS

Supports generation, topology, relative relationships and offline display - not ownership, exact area or approval.

AGENT-DESIGNED DATA

GeoJSON and metrics support conceptual zoning, capacity, networks, scenarios and phasing - not surveyed conditions.

ADMINISTRATIVE STATISTICS

Haidian, national and Beijing data calibrate problems - they do not establish corridor flows, siting or performance.

BACKGROUND CASES

Six institutional sites support mechanism comparison only - not Haidian performance analogies or delivery guarantees.

BACKGROUND OBSERVATIONS -> DESIGN ACTIONS

92 / 123

key laboratories / filed and launched large models

Prioritize standards, failure records, IP and product validation.

37

universities in one public service case

Create four clinic routes; do not infer equal needs or partnerships.

76.5%

green mode share in central Beijing

Prioritize active travel, but measure this corridor before expansion.

1456 / 239

health institutions / community centers or stations

Verify directory and refer; never diagnose or infer corridor capacity.

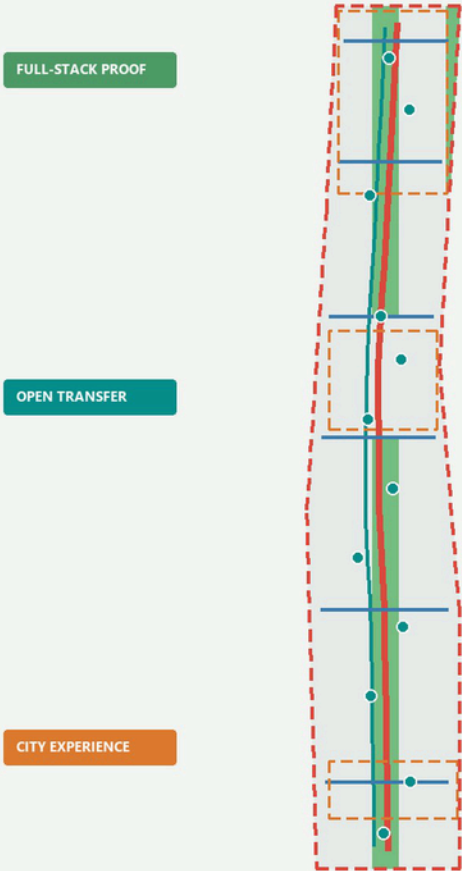
03 ONE SPINE, THREE STATIONS, TWO WINGS

A civic-validation loop across the Three-Level Scope Framework

01 PROOFLINE | OVERALL CONCEPT AND SPATIAL STRUCTURE

One civic spine | three Proof Stations | two service wings | five validation steps

JING-ZHANG PROOFLINE
PROVISIONAL | CONCEPT ONLY



CORE PROPOSITION

Urban AI should not merely display capability. Questions, tests, decisions, exits and contributions must be publicly visible.

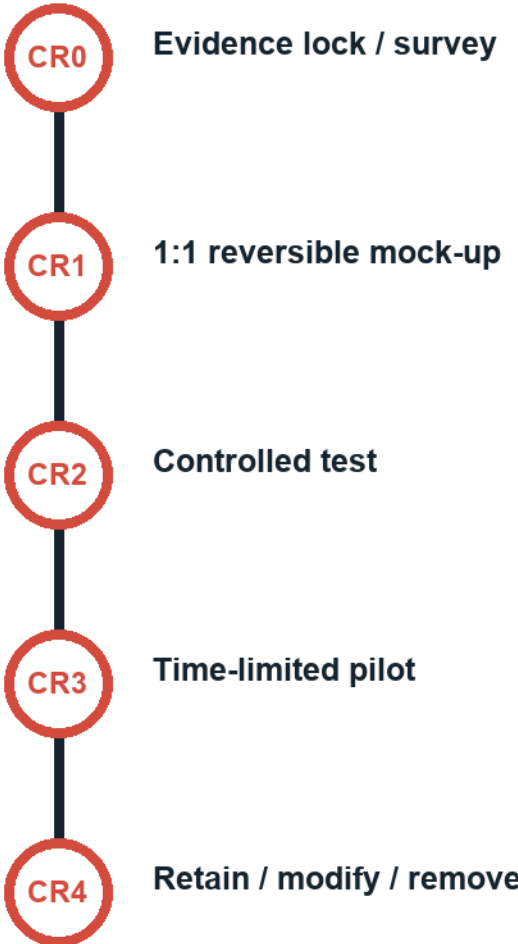
- CIVIC VALIDATION LOOP
- 01 DEFINE THE QUESTION
Residents, developers, companies and professional teams define it together.
 - 02 TEST IN PUBLIC
Run a minimum-scale trial at three Proof Stations and twelve scenario nodes.
 - 03 HUMAN DECISION
Accountable professionals decide and keep appeal and exit available.
 - 04 RECORD CONTRIBUTION
Archive traceable rules, failures, corrections and civic contributions.
 - 05 TRANSFER AND REUSE
Move validated components into products, places and operations through two wings.

PROVISIONAL BOUNDARY

The red dashed extent is a provisional constraint for concept generation and self-check. Recalculate every area, alignment and key-area form after official polygons.

ONE COMPONENT, ONE PROOF-MILE

Each component binds one location, public problem and its own evidence.



one pass ≠ line-wide pass ≠ construction approval

L1 ORDINARY-SERVICE BASE

works without a model

L2 REVERSIBLE TEST POCKET

authorized · removable · optional

L3 PUBLIC EVIDENCE RAIL

operator · negatives · removal

04 LAND USE AND SPATIAL STRUCTURE

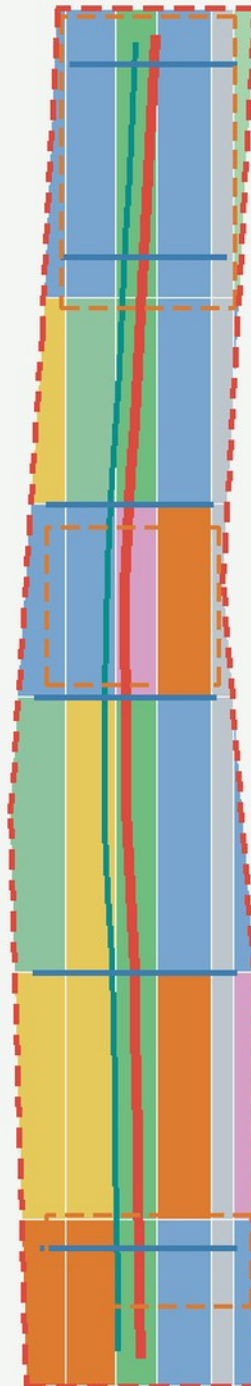
Complete conceptual coverage; no blank or statutory claim

02 COMPLETE LAND-USE PARTITION | THE SPINE IS NOT LEFTOVER SPACE

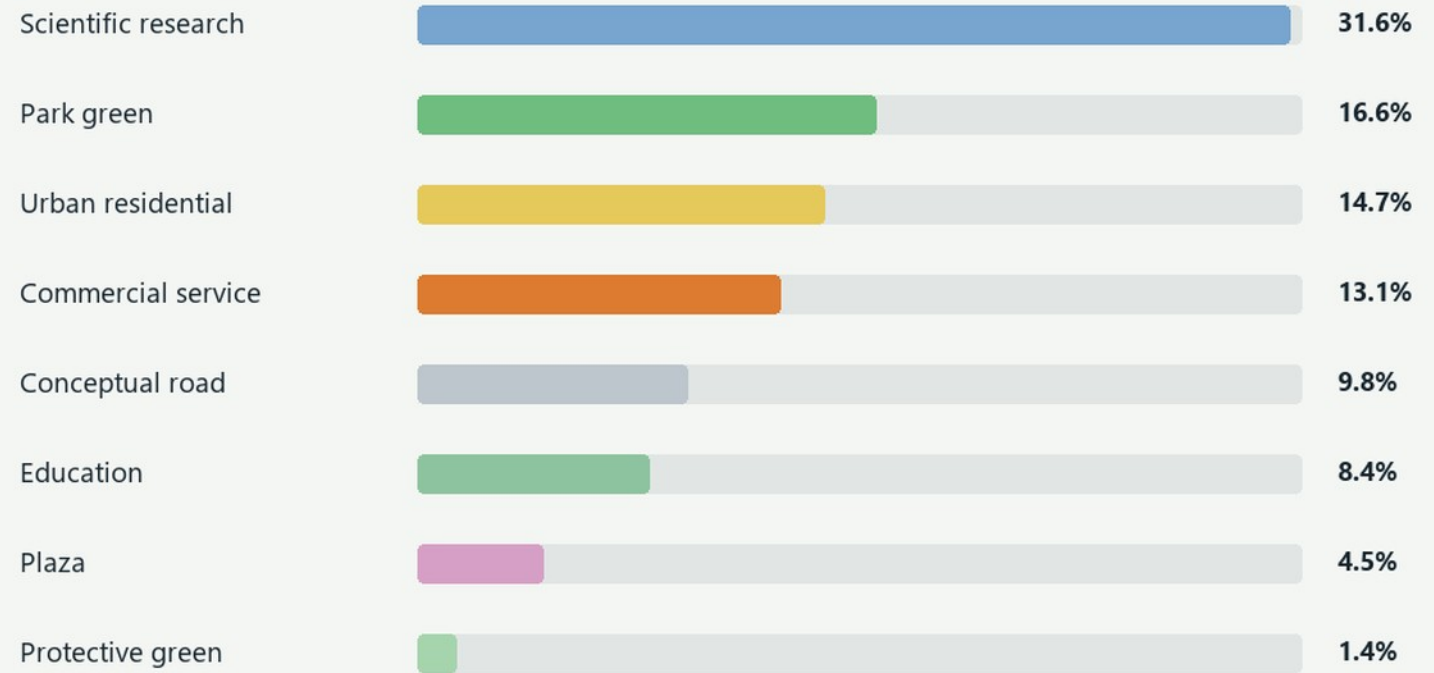
One shared grid | full coverage | conceptual use does not replace approved Regulatory Detailed Planning

JING-ZHANG PROOFLINE

PROVISIONAL | CONCEPT ONLY



FUNCTIONAL COMPOSITION



SPATIAL TRANSLATION

- 17%+ CONTINUOUS OPEN SPINE** Combine stormwater, shade, walking and civic validation along the railway direction.
 - RESEARCH NEXT TO EDUCATION** Share divisible proof, transfer and learning space while respecting campus ownership.
 - URBAN EXPERIENCE IN THE SOUTH** Place terminals, commerce and exchange at an opt-out interface with Human Review.

Data: geometry/land_use.geojson | EPSG:4548 recalculation | white seams are shared partition lines | red dash is provisional

05 DETAILED DESIGN OF THREE KEY AREAS

Full-stack proof, open transfer and opt-out urban experience

03 THREE PROOF STATIONS | ONE SPINE, THREE INNOVATION LOOPS

Key-area polygons are provisional | each station pairs civic value with an evidence gate and exit

JING-ZHANG PROOFLINE

PROVISIONAL | CONCEPT ONLY

01

Zhongzhiyuan

FULL-STACK PROOF STATION



- Model safety and standards sandbox
- Edge-computing carbon ledger
- Qinghe ecological innovation interface

EVIDENCE GATE BEFORE DETAIL

Heritage | waterway | transport | ownership | energy

02

Beijing AI Origin

OPEN TRANSFER STATION



- Open-source launch and contribution lineage
- Research transfer clinic
- Talent learning and everyday-life loop

EVIDENCE GATE BEFORE DETAIL

Campus | ownership | buildings | transit interface

03

Dazhongsi

CITY EXPERIENCE STATION



- Opt-out intelligent-terminal experience
- International demonstration and referral
- Four-quadrant walking connections

EVIDENCE GATE BEFORE DETAIL

Road | rail | utilities | commercial operations

06 TWELVE AI-ENABLED SCENARIOS

Every card has minimum data, a human reviewer, an exit and a stop condition

SCN-01

Model Safety Proving Ground

Zhongzhiyuan | industry test

Minimum necessary data | Human Review | immediate exit or appeal | explicit stop condition

SCN-02

Edge Computing Carbon Ledger

Zhongzhiyuan | industry test

Minimum necessary data | Human Review | immediate exit or appeal | explicit stop condition

SCN-03

Robot Right-of-Way Sandbox

Xiaoyue River | industry test

Minimum necessary data | Human Review | immediate exit or appeal | explicit stop condition

SCN-04

Open-Source Launch Hall

AI Origin | innovation

Minimum necessary data | Human Review | immediate exit or appeal | explicit stop condition

SCN-05

Research Transfer Clinic

AI Origin | enterprise service

Minimum necessary data | Human Review | immediate exit or appeal | explicit stop condition

SCN-06

Accessible Walking Companion

Proofline | public service

Minimum necessary data | Human Review | immediate exit or appeal | explicit stop condition

SCN-07

Maintenance Co-Review

Proofline | civic governance

Minimum necessary data | Human Review | immediate exit or appeal | explicit stop condition

SCN-08

Trustworthy AI Education

Near-campus | education

Minimum necessary data | Human Review | immediate exit or appeal | explicit stop condition

SCN-09

Health-Service Navigation

Community point | referral only

Minimum necessary data | Human Review | immediate exit or appeal | explicit stop condition

SCN-10

Legal-Service Referral

Community point | professional review

Minimum necessary data | Human Review | immediate exit or appeal | explicit stop condition

SCN-11

Opt-Out Terminal Experience

Dazhongsi | urban experience

Minimum necessary data | Human Review | immediate exit or appeal | explicit stop condition

SCN-12

Urban Memory Contribution

Proofline | culture

Minimum necessary data | Human Review | immediate exit or appeal | explicit stop condition

SHARED PROTOCOL Service user | minimum data | spatial boundary | human reviewer | exit / appeal | evaluation / stop

07

WALKING, CYCLING AND BLUE-GREEN PUBLIC SPACE

One continuous spine; expansion only after five corridor evidence gates

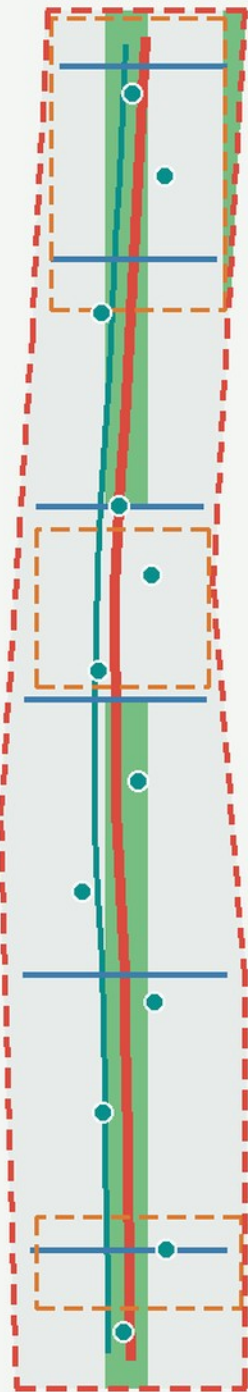
04

WALKING AND CYCLING x BLUE-GREEN PUBLIC SPACE | SIX CROSS-CONNECTIONS

Walking first | companion cycling | accessible continuity | transit referral | low-impact scenario nodes

JING-ZHANG PROOFLINE

PROVISIONAL | CONCEPT ONLY



Conceptual connection diagram, not a road boundary, rail alignment, bridge/tunnel scheme or engineering approval.

ONE CONTINUOUS CIVIC SPINE

The Jing-Zhang direction becomes a civic-validation spine. A companion cycling route separates speeds; six east-west interfaces connect communities, campuses, transit and parks into one everyday network.

PROOFLINE LENGTH

9.3 km

AI SCENARIO NODES

12

WALKING + CYCLING NETWORK

24.4 km

INDUSTRY VALIDATION

3

INCLUSIVE OPERATING PROTOCOL

- 01

DEVICE-FREE BY DEFAULT

Every service retains a staffed window, ordinary wayfinding and a non-smart alternative.
- 02

VISIBLE BOUNDARIES

Purpose, retention, responsible person, exit and appeal are readable on site.
- 03

LOW SPEED AND TIME SLOTS

Robot trials use speed limits, physical separation and on-site takeover.
- 04

FAILURES ENTER THE RECORD

Withdrawal reasons and negative results are archived alongside successes.

P01 Civic Proof Protocol EVIDENCE / STOP Legal, ethical, accessibility and public review	P02 Lightweight spine pilot EVIDENCE / STOP Heritage, ownership, fire and five transport baselines
P03 Zhongzhiyuan Safety Proof Garden EVIDENCE / STOP Energy, waterway, transport, data safety and human stop	P04 AI Origin Transfer Clinic EVIDENCE / STOP Ownership, safety, authorization and material governance
P05 Dazhongsi Opt-Out Experience Street EVIDENCE / STOP Transport baselines, fire, consumer rights and privacy	P06 Six accessible cross-connections EVIDENCE / STOP Continuity audit and representative-user review
P07 Three contribution nodes EVIDENCE / STOP Provenance, rights and dispute process	P08 Annual Proofline operations EVIDENCE / STOP Actor, budget, permit and safety responsibility

AREA PHASING



JING-ZHANG PROOFLINE · v1.6

CORE METRICS × ONE COMPONENT, ONE PROOF-MILE

ONE DATASET · ONE MECHANISM · ONE FORMAL STORY

PROVISIONAL OVERALL AREA
1141.3 ha
EPSG:4548 | not official redline

CONCEPTUAL GREEN RATIO
18.0%
metrics.json | not statutory

PUBLIC SPACE RATIO
4.5%
plaza polygons only

CONCEPTUAL BCR
15.4%
capacity test | statutory FAR unknown

AI / INDUSTRY SCENARIOS
12 / 3
8 conceptual action packages

EACH COMPONENT EARNS ITS OWN MILE

one pass ≠ line-wide pass ≠ construction approval



ONE CONSTRUCTION BAY · THREE LAYERS

The ordinary route works; the test retreats into a pocket; evidence returns on site.

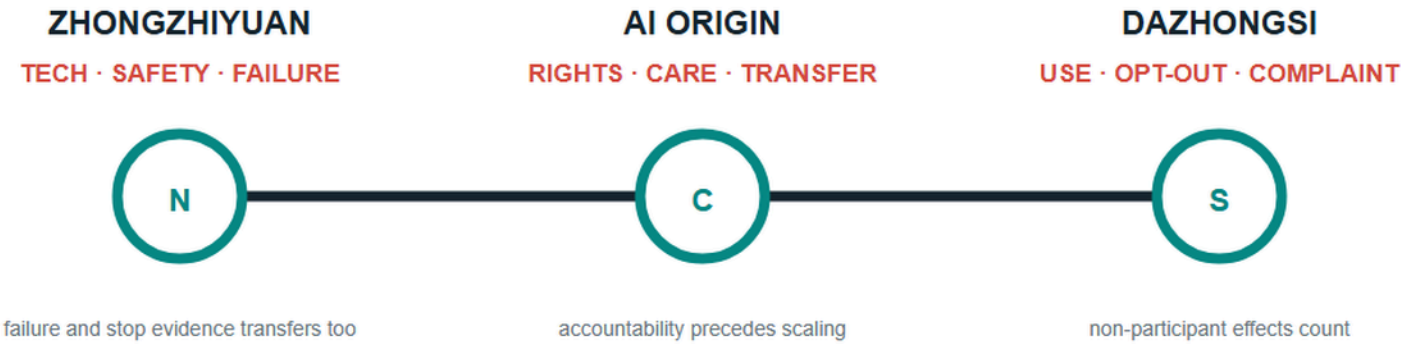
L1 ORDINARY-SERVICE BASE
Passage / wayfinding / rest / staffed help without model, power or network

L2 REVERSIBLE TEST POCKET
Authorized, removable, outside clear width, never compulsory

L3 PUBLIC EVIDENCE RAIL
Component / gate / operator / baseline / negative findings / stop / removal

THREE PROOF STATIONS · THREE HANDOFFS

A rejectable component record—not device count—moves along the line.



BOUNDARY: No new geometry, quantity, dimension, strength, alignment, schedule or cost; CR passes never replace professional or statutory approval.

Five metrics match metrics.json; Proof-Mile is a conceptual delivery mechanism; CR passes are not construction approval.

10

RISK, COPYRIGHT AND COMPLIANCE

Review-ready is not approved, funded, engineered or officially bounded

PROVISIONAL BOUNDARY

Replace official polygons first, then regenerate every design layer, metric and display.

BUILDING DECISIONS

Capacity prototypes are not surveyed buildings and mark no demolition target.

SPATIAL MISALLOCATION

Administrative statistics never establish corridor demand or project performance.

AI GOVERNANCE

Minimum data, Human Review, non-digital alternatives, appeal, exit and a stop date are mandatory.

STATUTORY MISREADING

Land use and capacity are low-confidence concepts; statutory FAR remains unknown.

ENGINEERING ALIGNMENT

Mobility lines express connection intent and require road, rail, utility and accessibility review.

PLATFORM DATA

Personal traces and order records are prohibited; aggregate data requires permission and bias disclosure.

IMPLEMENTATION CLAIM

Actors, permits, budget, procurement and responsibility remain to be established.

ORIGINAL AND OFFLINE

Narrative, GeoJSON, JSON, HTML, logo, evidence figures and PDFs are original submission outputs. The concept image is presentation only. Offline HTML loads no CDN, remote map, font, script, iframe, form, API or tracker.